

Anderson-Higgs Mass of Magnons in Superconductor-Ferromagnet-Superconductor Systems

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The Anderson-Higgs mechanism of mass generation is a generic concept in high-energy and condensed-matter physics. It shows up through the Meissner effect providing the expulsion of static and low-frequency magnetic fields from superconductors. However, it does not affect propagating electromagnetic waves with a spectrum gap determined by the plasma frequency, which is too large to be sensitive to the superconducting transition. Here we demonstrate the spectroscopic manifestation of the Anderson-Higgs mass, showing that it determines the spectrum gap of magnons in superconductor/ferromagnet/superconductor multilayers. Moreover, we show that this effect has been observed in recent experiments as a spontaneous ferromagnetic resonance frequency shift in such systems. Our theory explains many unusual experimental features and suggests effective controls over the magnon spectrum with tunable spectral gap and group-velocity reversal. These findings pave the way to a wide range of advanced functionalities for possible applications in magnonics.

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Introduction—The concept of spontaneous symmetry breaking and mass generation in local gauge theories plays central roles both in particle physics and in condensed-matter systems. The Anderson-Higgs (AH) mechanism [1] of mass generation resulting from the spontaneous breaking of local gauge symmetries takes place both in charged superconductors [2,3] and in the Standard Model [4–6]. However there is an important conceptual difference between these cases. In high-energy theories the vector boson masses are determined solely by the AH mechanism. On the contrary, in metals the propagating photons have an energy gap equal to the plasma frequency already in the normal state as a result of interactions [3]. It is determined by the total density of electrons and therefore is not sensitive to the superconducting transition. Rather, superconductivity shows up in the Meissner effect, that is, the modification of the photon decay length at frequencies much smaller than the plasma one. Therefore when it comes to condensed-matter systems it is fair to say that the AH mass of propagating excitations has never been observed.

In this Letter we demonstrate that the energy gap due to the AH mechanism can be directly observed for magnons in hybrid systems consisting of coupled metallic ferromagnet (*F*) and superconducting (*S*) layers. The electromagnetic interaction between the dynamical magnetization and the superconducting condensate leads to the

spontaneous generation of magnon spectral gap in the absence of any magnetic anisotropy. We argue that this phenomenon has been detected in recent experiments as a spontaneous shift of the ferromagnetic resonance (FMR) frequency controlled by the superconducting order parameter in *S/F/S* systems, with Nb and Py materials as *S* and *F*, respectively [7–9]. Thus the superconductivity can be used for the manipulation of magnon states for spin-wave logic [10] and magnonics [11–13]. Our proposal provides a step toward applications of superconducting spintronics [14–21], superconducting magnonics [7–9,22–31], and on-chip cavity magnonics [32–36] applications.

Qualitative picture—We start with the qualitative argument. Consider the *F* slab with in-plane magnetization in the absence of external field and any in-plane anisotropies shown in Fig. 1(a). The magnetization rotation in the *xy* plane $\mathbf{M} = M_0(\cos \theta, \sin \theta, 0)$ is a massless mode, which means that the magnetic energy is degenerate by the direction θ , which plays the role of the *U*(1) gauge field. As a result, the magnon spectrum is also massless having no energy gap. Consider, for example, magnetostatic surface spin waves in a ferromagnet slab in the Damon-Eshbach geometry [37] when the external magnetic field $\mathbf{H}_0$ is aligned in the *xy* plane $\mathbf{H}_0 \perp \mathbf{z}$ and perpendicular to the magnon wave vector $\mathbf{q} \perp \mathbf{H}_0$. Magnon dispersion is given by [38]

$$\Omega_{\text{mag}}(q) = \sqrt{\Omega_K^2 + \frac{\Omega_M^2}{4}(1 - e^{-2qd_F})}, \quad (1)$$

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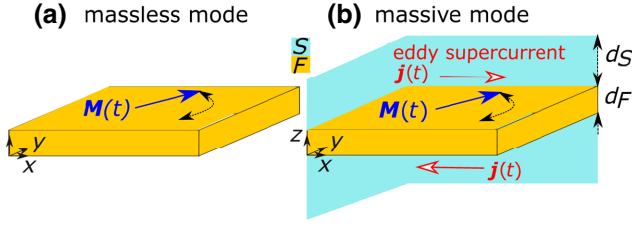


FIG. 1. (a) In the F slab the in-plane magnetization rotation $\mathbf{M}(t)$ is a zero-mass mode. (b) In the $S/F/S$ system the in-plane magnetization rotation $\mathbf{M}(t)$ is a massive mode because of the eddy supercurrents $\mathbf{j}(t)$.

where $\Omega_K = \gamma \sqrt{H_0(H_0 + 4\pi M_0)}$ is the Kittel frequency [39] of the FMR in F slab, $\Omega_M = 4\pi \gamma M_0$, and M_0 is the magnetization of F . For vanishing external field $H_0 \rightarrow 0$ the minimal frequency vanishes $\Omega_{\text{mag}}(q=0) \rightarrow 0$ corresponding to the massless mode.

The situation changes in the presence of superconductors in contact with ferromagnetic film as shown in Fig. 1(b). According to the Faraday law, time-dependent magnetization components $\mathbf{M}(t) = \mathbf{M}_0 + \mathbf{M}_\omega e^{i\omega t}$ generate a curly electric field outside the F film $\mathbf{E}_\omega = \pm(2\pi i\omega/c)d_F \mathbf{z} \times \mathbf{M}_\omega$ with upper sign for $z > d_F/2$ and lower sign for $z < -d_F/2$. This electric field induces eddy currents in the adjacent metallic layers [40,41]. In the present case of S layers these are the eddy supercurrents $\mathbf{j}_\omega = \sigma_S(\omega)\mathbf{E}_\omega$, where the frequency-dependent conductivity of S is given by [42]

$$\sigma_S(\omega) = \frac{c^2}{4\pi i \lambda^2 \omega}. \quad (2)$$

Here λ is the London penetration length. We assume the frequency to be much smaller than the superconducting gap so that the normal component contribution can be neglected. As a result, eddy supercurrents are given by $\mathbf{j}_\omega = \pm(cd_F/\lambda^2)\mathbf{M}_\omega \times \mathbf{z}$. Remarkably, the prefactor coefficient here does not depend on frequency. Thus, assuming for simplicity the thickness of S layers to be small, $d_S \ll \lambda$, we get the contribution of the kinetic energy of the eddy supercurrents contributing to the action describing the magnetic degrees of freedom normalized by the F thickness:

$$S_m = -2\pi M_0^2 \frac{d_S d_F}{\lambda^2} \int dt d\theta \theta^2, \quad (3)$$

where the angle θ determines the deviation of $\mathbf{M}(t)$ from its stationary direction. Equation (3) is the AH mass-generating term for magnons.

The Goldstone mode that is “absorbed” by the $U(1)$ gauge field θ to get Eq. (3) is the Josephson phase difference φ between the upper and lower S layers in Fig. 1(b). Because of the strong exchange field and large F layer

thickness the Josephson energy is negligible. In general the action Eq. (3) has the gauge-invariant form obtained by replacing $\theta \rightarrow \theta + L_m \nabla_x \varphi$, where $L_m = \phi_0/(4\pi^2 d_F M_0)$ and ϕ_0 is the flux quantum. The time-dependent Josephson phase difference $\varphi(t)$ generates a voltage $V(t) = (\phi_0/2\pi)\partial_t \varphi$ between upper and lower S layers. In the case of metallic F with conductivity σ_F due to the skin effect this ac voltage V and hence φ are localized near the edges of the $S/F/S$ structure at the skin length $\text{Im}l_{\text{sk}}$, where $l_{\text{sk}} = c\sqrt{1/4\pi i\omega\sigma_F}$. Away from this shell $V = 0$ and $\varphi = 0$ so that the action has the form of Eq. (3).

Model—To put the above qualitative reasoning on the solid ground we consider below the exact calculation of the FMR fundamental mode and magnetostatic waves in $S/F/S$ structures. Our starting equations for the frequency components of magnetization $\mathbf{M}_\omega$, current $\mathbf{j}_\omega$, and magnetic $\mathbf{H}_\omega$ and electric $\mathbf{E}_\omega$ fields are

$$\text{LLG: } i\omega \mathbf{M}_\omega = \gamma(\mathbf{B}_0 \times \mathbf{M}_\omega + \mathbf{B}_\omega \times \mathbf{M}_0), \quad (4)$$

$$\text{in metal: } \nabla \times (\sigma^{-1} \nabla \times \mathbf{H}_\omega) + \frac{4\pi i\omega}{c^2} \mathbf{B}_\omega = 0, \quad (5)$$

$$\text{in insulator: } \nabla \times \mathbf{H}_\omega = 0. \quad (6)$$

This system consists of the Landau-Lifshitz-Gilbert (LLG) equation [Eq. (4)], where $\mathbf{B}_0 = \mathbf{H}_0 + 4\pi \mathbf{M}_0$ is the stationary magnetic field. Length scales are assumed larger than the exchange length and gradient terms in LLG are neglected. Maxwell equations in metal taken in the usual quasistatic approximation [43] yield Eq. (5). The conductivity is $\sigma_F = \text{const}$ in F while in S it is given by Eq. (2). In a vacuum or insulator where $\sigma = 0$ we neglect the displacement current as well and use the Maxwell Eq. (6), which yields that $\mathbf{H}_\omega$ is a potential field. This approximation is valid provided the lateral size along x is much smaller than the wavelength at the given frequency. For $\omega \sim 10$ GHz, it is $c/\omega \sim 1$ cm.

Fundamental FMR mode—The fundamental FMR mode is then given by the eigenmode of the system of Eq. (5) where all fields depend only on z . We assume the lateral sizes of the $S/F/S$ structure to be sufficiently large to neglect edge effects. Time-dependent magnetization components are $\mathbf{M}_\omega = (0, M_y, M_z)$. The B_z component is continuous so that $B_z = 0$ and therefore $H_z = -4\pi M_z$ everywhere. Under such conditions solving the LLG equation [Eq. (4)] in terms of H_y yields $B_y = (\Omega_B^2 - \omega^2)/(\Omega_K^2 - \omega^2)H_y$, where $\Omega_B = \gamma B_0$. The field $H_y \neq 0$ is characteristic for metallic ferromagnets [40] and provides the detuning of FMR frequency from the Kittel one Ω_K . Substituting the found B_y into Eq. (5) we get

$$\text{in } F: \quad \nabla_z^2 H_y - l_F^{-2} H_y = 0, \quad (7)$$

$$\text{in } S: \quad \nabla_z^2 H_y - \lambda^{-2} H_y = 0. \quad (8)$$

Here the renormalized skin depth length in F is determined by

$$l_F = l_{sk} \sqrt{(\Omega_K^2 - \omega^2)/(\Omega_B^2 - \omega^2)}. \quad (9)$$

Boundary conditions at F/S interfaces follow from Eq. (5):

$$H_y|_F = H_y|_S, \quad (10)$$

$$\nabla_z H_y|_F = \frac{\sigma_F}{\sigma_S} \nabla_z H_y|_S. \quad (11)$$

The boundary condition at the S/I and F/I interfaces, where I stands for the insulating layer or vacuum, is $H_y|_I = 0$. Indeed, in the considered case when all fields do not depend on y we obtain from Eq. (6) that $\nabla \times \mathbf{H}_\omega = \mathbf{z} \nabla_x H_y - \mathbf{x} \nabla_z H_y - \mathbf{y} \nabla_x H_z = 0$, so that $\nabla_x H_y = 0$. Integrating this equation along x from any point in the I layer to infinity outside the $S/F/S$ structure, where $H_y(x = \infty) = 0$, we get that $H_y = 0$ everywhere in the I region. The eigenmodes of the system Eqs. (7) and (8) with boundary conditions determine the FMR frequency.

We start with the simplest symmetric case when $d_{S1} = d_{S2} = d_S$ in Fig. 2(a). The solution of Eqs. (7) and (8) is symmetric about the middle of F , $z = 0$:

$$\text{in } F: \quad H_y(z) = \tilde{H}_F \cosh(z/l_F), \quad (12)$$

$$\text{in } S: \quad H_y(z) = \tilde{H}_S \sinh[(z - d_S - d_F/2)/\lambda]. \quad (13)$$

Matching solutions at $z = d_F/2$ using Eqs. (10) and (11) and using the identity $\sigma_S \lambda^2 / \sigma_F l_{sk}^2 = 1$ we get the general equation determining the FMR frequency:

$$\tanh(d_S/\lambda) \tanh(d_F/2l_F) = -\lambda l_F / l_{sk}^2. \quad (14)$$

For frequencies $\omega < 100$ GHz and conductivity [44] of Py of $\sigma_F \sim 10^6 \Omega^{-1} \text{m}^{-1}$, the nonmagnetic metal skin depth is $l_{sk} > 10 \mu\text{m}$. Thus for realistic F layer thickness $d_F < 1 \mu\text{m}$, the condition $d_F \ll l_{sk}$ is satisfied. If the frequency is far from the Kittel mode we have $l_F \sim l_{sk}$ and based on

the above estimations can expand $\tanh(d_F/2l_F) \approx d_F/2l_F$. Thus we obtain from Eq. (14) the explicit expression for the FMR frequency:

$$\Omega_{\text{FMR}} = \gamma \sqrt{\frac{B_0^2 (d_F/2\lambda) \tanh(d_S/\lambda) + B_0 H_0}{(d_F/2\lambda) \tanh(d_S/\lambda) + 1}}. \quad (15)$$

The general Eq. (15) is qualitatively different from the Kittel expression [39] with generalized demagnetizing factors [45]. For small external field $H_0 \ll B_0$ and F layer thickness $d_F \ll \lambda \tanh(d_S/\lambda)$, Eq. (15) reduces to the Kittel expression [39] with artificial anisotropy energy $E_a = K_a M_y^2$, where $K_a = 2\pi (d_F/\lambda) \tanh(d_S/\lambda)$. In the case $d_S \ll \lambda$ we get $E_a = 2\pi M_0^2 (d_F d_S / \lambda^2) \theta^2$ in agreement with Eq. (3). As shown below the AH mass of magnons Ω_{AH} is given by the FMR frequency shift $\Omega_{\text{AH}} = \Omega_{\text{FMR}}(H_0 = 0)$ so that

$$\Omega_{\text{AH}} = 4\pi \gamma M_0 \sqrt{\frac{(d_F/2\lambda) \tanh(d_S/\lambda)}{(d_F/2\lambda) \tanh(d_S/\lambda) + 1}}. \quad (16)$$

At larger F thickness $d_F \gg \lambda \coth(d_S/\lambda)$ the AH magnon mass Eq. (15) saturates at $\Omega_{\text{AH}} = 4\pi \gamma M_0$. Recently the tendency to such saturation has been observed experimentally [46] in Nb/Py/Nb multilayers. Taking into account the temperature dependence of $\lambda(T)$ calculated within the model of diffusive superconductor, we get the behavior $\Omega_{\text{AH}}(d_F, T)$ shown in Figs. 3(a) and 3(b). The fixed parameters relevant to the Nb/Py/Nb experimental systems [7–9] are $d_S = 150$ nm, $\lambda(T=0) = 80$ nm, $4\pi M_0 = 1.06$ T. From Fig. 3(b) it is clear that the temperature dependencies of $\Omega_{\text{AH}}(T)$ are rather close to that of the inverse London length in $\lambda^{-1}(T)$ in accordance with experiments [9,46].

As is clear from Eq. (16) and Fig. 3(a), the scale of Ω_{AH} is determined by $\Omega_M = 4\pi \gamma M_0$, which for Py is $\Omega_M \approx 30$ GHz. Thus, the observed FMR frequency shifts [9,46] of the order of 10 GHz are explained by theory. To be more convincing, in Fig. 3(c) we show the direct numerical comparison of theoretical dependencies $f_{\text{FMR}}(H_0) = \Omega_{\text{FMR}}(H_0)/2\pi$ given by Eq. (15) with experimental data presented in Fig. 2(c) of Ref. [9]. Theoretical curves and experimental data are shown by solid and dashed lines, respectively. As one can see, the present theory provides very accurate fits of the experimental $f_{\text{FMR}}(H_0)$ dependencies for all temperatures. All parameters except d_S are chosen exactly the same as in the experiment [9], namely $T_c = 9$ K, $\lambda(T=0) = 80$ nm, $d_F = 19$ nm, $4\pi M_0 = 1.06$ T, so that $\Omega_M = 31.3$ GHz. The only parameter we adjust to get most accurate fits in Fig. 3(c) is $d_S = 80$ nm, which is slightly smaller than in the experiment: $d_S = 110$ nm. This is natural given that there is a superconductivity suppression near the S/F interfaces at the scale of coherence length [47], which is $\xi_0 \sim 10\text{--}40$ nm in Nb.

Let us consider a more general case when S layer thicknesses are different: $d_{S1} \neq d_{S2}$. A slightly more

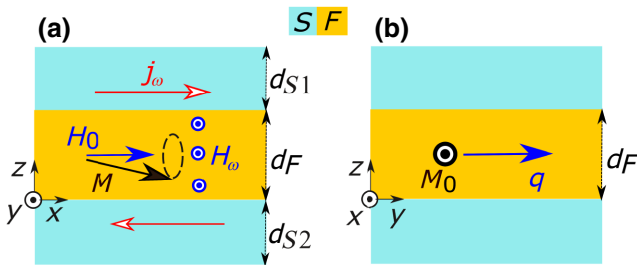


FIG. 2. (a) Ferromagnetic resonance in $S/F/S$ system. Distributions of loop eddy currents $\mathbf{j}_\omega$ and the magnetic field that they generate $\mathbf{H}_\omega$ are shown schematically. (b) The magneto-static mode in $S/F/S$ structure in Danone-Eschbach geometry with wave vector $\mathbf{q} \perp \mathbf{M}_0$.

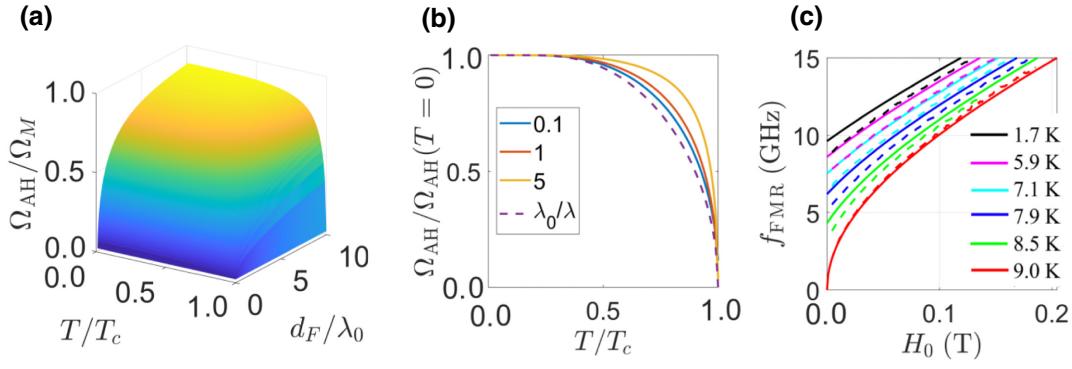


FIG. 3. (a),(b) Anderson-Higgs mass of magnons Ω_{AH} given by Eq. (16). (a) $\Omega_{\text{AH}}(T, d_F)$ normalized by $\Omega_M = 4\pi\gamma M_0$, showing saturation at $T \ll T_c$ and $d_F \gg \lambda$. (b) $\Omega_{\text{AH}}(T)$ for $d_F/\lambda_0 = 0.1, 1, 5$ shown by solid lines compared with $\lambda_0/\lambda(T)$ shown by the dashed line, where $\lambda_0 = \lambda(T=0)$. (c) Comparison of $f_{\text{FMR}} = \Omega_{\text{FMR}}/2\pi$ given by Eq. (15) shown by solid lines with experimental data from Fig. 2(c) of Ref. [9] shown by dashed lines. Parameters are detailed in the text.

involved calculation [48] yields

$$\Omega_{\text{FMR}} = \gamma \sqrt{\frac{B_0^2(d_F/\lambda)[\coth(d_{S1}/\lambda) + \coth(d_{S2}/\lambda)]^{-1} + B_0 H_0}{(d_F/\lambda)[\coth(d_{S1}/\lambda) + \coth(d_{S2}/\lambda)]^{-1} + 1}}. \quad (17)$$

This expression describes how FMR shift disappears together with the thickness of one of the S layers with the transition from $S/F/S$ to S/F structure. If, for example, $d_{S1} \rightarrow 0$, Eq. (17) yields the usual Kittel frequency $\Omega_{\text{FMR}} \rightarrow \Omega_K = \gamma\sqrt{H_0 B_0}$. Thus, the FMR frequency of the S/F structure equals Ω_K .

In the considered approximation of the structure length along x being much shorter than the wavelength, the FMR mode in S/F structure is equivalent to that in $S/F/I/S$ structure, where I is the insulator. This is because there is the same boundary condition $H_y = 0$ at the F/I or F/vacuum interface. This conclusion explains experiments [7–9] where seemingly enigmatic suppression of Ω_{AH} by the I layer has been observed. This phenomenon has an electromagnetic origin not connected with the influence of the spin-triplet proximity effect induced by magnetic precession [7,49]. Similarly, under the considered condition that the lateral size along x is much smaller than c/ω , superconductivity has no effect on the FMR mode in $S/FI/S$ structures. Here FI stands for the ferromagnetic insulator, e.g. the quite popular material yttrium iron garnet (YIG) [50]. In this case, $H_y = 0$ everywhere so that no currents are generated in the adjacent S layers. However, superconductivity affects the dispersion of spin waves in such setups [51].

Magnetostatic magnons in $S/F/S$ system—Now let us turn to the spin waves in the $S/F/S$ system in the geometry of Fig. 2(b) with $\mathbf{M}_0 \parallel \mathbf{x}$ searching the solution in the form where all fields are proportional to $e^{iq_y y}$. In this case the calculation becomes more involved as compared with

FMR due to the presence of both B_y and B_z components. From Eq. (5) (see the detailed derivation in Ref. [48]), we get

$$\text{in } F: \quad \nabla_z^2 H_y - (l_F^{-2} + q^2) H_y = 0, \quad (18)$$

$$\text{in } S: \quad \nabla_z^2 H_y - (\lambda^{-2} + q^2) H_y = 0, \quad (19)$$

supplemented by the boundary conditions at S/F interfaces, which consist of continuity Eq. (10) and

$$(\nabla_z H_y - iq H_z)|_F = \frac{\sigma_F}{\sigma_S} (\nabla_z H_y - iq H_z)|_S. \quad (20)$$

We consider the range of momenta $ql_{\text{sk}} \gg 1$, which is experimentally relevant given that $l_{\text{sk}} \sim 10 \mu\text{m}$. For simplicity we assume $d_{S1,2} \gg \lambda$. In this approximation, it is possible to find magnon dispersion solving Eqs. (18)–(20) analytically [48]:

$$\Omega_{\text{mag}}(q) = \sqrt{\frac{(\Omega_M + a\Omega_0)^2 - (\Omega_M + b\Omega_0)^2 e^{-2qd_F}}{a^2 - b^2 e^{-2qd_F}}}, \quad (21)$$

where $\Omega_0 = \gamma H_0$, $a = 1 + q\lambda/\sqrt{1 + q^2\lambda^2}$, and $b = 1 - q\lambda/\sqrt{1 + (q\lambda)^2}$. Equation (21) yields two limiting cases. First, at large momenta $q\lambda \gg 1$ it reduces to the usual surface spin-wave dispersion Eq. (1). The reason is that the fields outside F decay at scales $q^{-1} \ll \lambda$ and therefore cannot excite eddy supercurrents. Second, at $q = 0$ the magnon frequency Eq. (21) coincides with the FMR frequency Eq. (15). In the absence of external field $H_0 = 0$ it yields the AH magnon mass given by Eq. (16), $\Omega_{\text{mag}}(q = 0, H_0 = 0) = \Omega_{\text{AH}}$, taken in the limit $d_S \gg \lambda$.

Comparing $\Omega_{\text{mag}}(q = 0) = \Omega_{\text{AH}}$ and the asymptotic $\Omega_{\text{mag}}(q \rightarrow \infty) = \Omega_M/2$ we note that there are two qualitatively different cases. For $\Omega_{\text{AH}} < \Omega_M/2$, the dependence $\Omega_{\text{mag}}(q)$ is monotonically increasing, which is usual for

surface spin waves [37,52,53]. Such waves are “forward” ones since they have parallel group velocity and wave vector $v_g = \partial\Omega_{\text{mag}}/\partial q > 0$. On the other hand, for $\Omega_{\text{AH}} > \Omega_M/2$, the dependence $\Omega_{\text{mag}}(q)$ first becomes nonmonotonic and then at even larger Ω_{AH} it is monotonically decreasing. This regime corresponds to the “backward” wave $v_g < 0$ with antiparallel group velocity and wave vector. Provided that d_F is sufficiently large, this transition in magnon spectrum occurs at a certain threshold temperature determined by $\Omega_{\text{AH}}(T) = \Omega_M/2$. Shown in Fig. 4 are the two characteristic cases of $d_F = 0.5\lambda_0$ [Fig. 4(a)] when waves are “forward” for all temperatures and $d_F = 3\lambda_0$ [Fig. 4(b)] when there is a crossover from “forward” to “backward” regime at $T \approx 0.98T_c$. At this temperature there is almost flat magnon dispersion as shown in Fig. 4(d) by the red thick curve.

Summary—To summarize, we demonstrate that there exists an energy gap of magnons in $S/F/S$ systems, which appears spontaneously in the absence of external magnetic field or magnetic anisotropy. This magnon gap is interpreted as AH mass since it emerges upon the gauge-symmetry-breaking phase transition into the superconducting state. Previously the AH mechanism in condensed-matter systems has been commonly associated

with the Meissner effect responsible for the expulsion of electromagnetic fields from superconductors at frequencies substantially smaller than the plasma frequency threshold for propagating waves in metals. Our results show that AH mass can also be observed in a qualitatively different way by *spectroscopic* means through the modification of propagating magnon dispersion. Moreover, we show that AH mass has already been observed in a series of recent experiments [7–9,46,54] on FMR frequency shift in Nb/Py/Nb structures. The nonzero FMR frequency at $H_0 = 0$ has been detected and remained enigmatic until now. The present work establishes that this frequency shift is precisely the AH mass of magnons and explains all the unusual features observed experimentally. The derived Eq. (15) provides accurate fits of experimental data [9].

Proposed controls of spin waves with the help of superconductivity pave the way to various functionalities. For example, it is possible to localize magnons and fabricate magnon crystals by patterning one of the S layers. Alternatively, one can use light to modulate the order parameter and the AH magnon mass. It enables the creation of optically controlled magnon resonators and crystals. Moving this resonator along the F film and implementing magnon transport will be possible by shifting the focused light spot.

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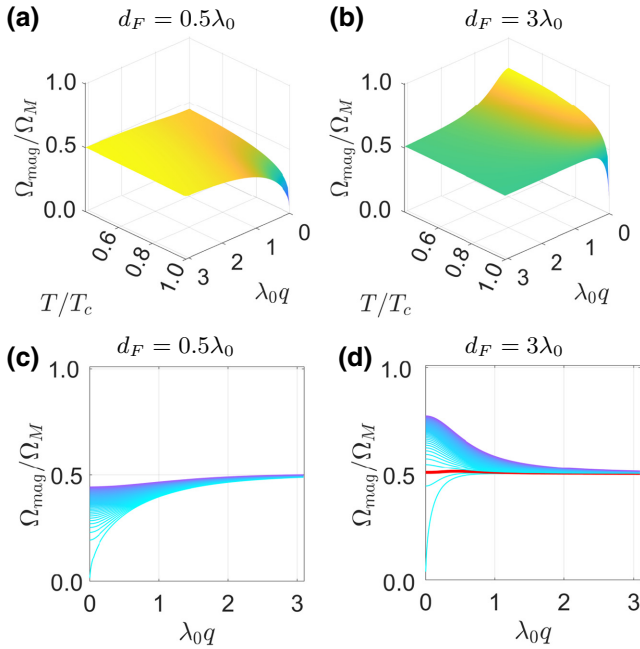


FIG. 4. The evolution of dispersion $\Omega_{\text{mag}}(q)$ of surface spin waves in $S/F/S$ system at different temperatures at $H_0 = 0$. (a) $d_F = 0.5\lambda_0$ and (b) $d_F = 3\lambda_0$. In the latter case the dispersion changes from “forward” $v_g > 0$ to “backward” $v_g < 0$ with decreasing temperature. (c),(d) The set of curves $\Omega_{\text{mag}}(q)$ for different temperatures starting from $T = T_c$ (lower curve) to $T = 0.5T_c$ (upper curve) with a step $0.01T_c$. The almost flat magnon spectrum shown by the red thick curve in (d) corresponds to the threshold at $T = 0.98T_c$.

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